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AI-Driven Time Series Models for Rainfall Prediction: A Machine Learning Approach

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ABSTRACT

Rainfall variability plays a critical role in agriculture, water resources, and disaster management in India. This study focuses on the usage of a SARIMAX model to analyze and predict rainfall patterns while accounting for both non-seasonal and seasonal components. The model adequately captures seasonal variations, illustrating the relevance of incorporating seasonal factors into forecasting. However, the non-seasonal components did not considerably improve the model's performance. Diagnostic tests demonstrate that the model handles autocorrelation satisfactorily; however, the residuals exhibit irregularities, as evidenced by skewness and large tails. While the model has fair prediction accuracy, the findings show areas for development, particularly in fine-tuning non-seasonal dynamics and eliminating residual abnormalities. This research emphasizes the importance of seasonality in rainfall forecasting and lays the framework for future model improvements. The study provides valuable information on rainfall patterns, supporting better planning and management

Keywords: Rainfall variability, Climate patterns, Time series analysis, Seasonal decomposition, ARIMA, SARIMAX.

I INTRODUCTION

Rainfall is a critical component of India's economy, deeply influencing agriculture, water management, and disaster preparedness. As a predominantly agrarian country, Indian agriculture heavily depends on the monsoon season to ensure consistent crop production, which is directly tied to food security and the livelihoods of millions. Beyond agriculture, rainfall patterns affect the management of water resources, affecting urban and rural water supply, the generation of hydroelectric power, and the health of natural ecosystems. Additionally, with climate change causing a rise in extreme weather events, understanding rainfall patterns has become increasingly important for disaster response and risk mitigation.

India's unique geographical diversity creates significant variations in rainfall across regions. Some areas experience excessive rainfall, leading to floods, while others face prolonged droughts, creating challenges for policymakers and communities alike. Analyzing rainfall trends at a district level allows for a more detailed understanding of these disparities, enabling tailored strategies to address local needs. Such fine-grained analysis can be invaluable for ensuring efficient irrigation, better reservoir management, and preparedness for extreme weather conditions.

Ultimately, understanding rainfall trends is not just about weather patterns but also about improving lives and livelihoods. Farmers can use this information to decide what crops to grow and when to plant, while policymakers can design effective disaster management systems. Urban planners can create better drainage infrastructure, and environmentalists can advocate for sustainable practices based on real data. This research strives to connect the dots between data analysis and actionable solutions, helping India adapt to its ever-evolving climatic challenges.

II RELATED WORK

Rahman and Lateh (2017) analyzed climate trends in Bangladesh between 1971 and 2010 using data from 34 meteorological stations. Their study observed a 0.20°C temperature rise per decade and a 7.13 mm annual increase in rainfall, though fluctuations were noted in the pre- and post-monsoon seasons [1].

Graham and Mishra (2022) applied SARIMA models to forecast rainfall in Allahabad using 31 years of data. They found the model effective but noted accuracy issues

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due to missing values, suggesting better data handling methods for improvement [2].

Gajbhiye et al. (2022) studied rainfall patterns in the Sindh River basin from 1901 to 2002, identifying seasonal variations and declining trends in certain months. Their findings underscored the importance of effective water resource planning to address changing rainfall patterns [3].

Longobardi and Villani (2018) examined precipitation changes in Italy (1918-1999) and reported a 135 mm rainfall decline over 50 years, with a slight increase during summer. They emphasized the need for geostatistical techniques to assess localized climate variations [4].

Naill and Momani (2022) used ARIMA models to predict rainfall in Jordan (1922-1999) and demonstrated its usefulness in hydrology. However, they noted challenges in forecasting extreme rainfall and recommended intervention time series analysis to enhance accuracy [5].

Ahmad et al. (2023) explored deep learning applications in cloud, edge, fog, and IoT computing. Their study highlighted the ability of deep learning to process large-scale data efficiently, particularly for real-time analytics. They emphasized the need for hybrid AI models to improve scalability and computational efficiency in data-driven applications[20].

Collectively, these studies highlight the effectiveness of ARIMA and SARIMA models in rainfall forecasting while also pointing out their limitations, particularly in handling missing data and extreme weather patterns. Future research should investigate machine learning approaches like LSTMs and XGBoost, which have shown promise in improving predictive accuracy and adaptability in climate modeling.

III METHODS AND MATERIALS

Several studies have found that the applicable models (e.g., ARIMA, SARIMA) may not fully reflect the complexity of rainfall patterns, particularly when predicting extreme events such as peak rainfall. Non-linear trends or seasonal variability may be more difficult for models such as ARIMA to handle.

A. Dataset Description

This study uses district-wise daily rainfall data for India, containing records of rainfall measurements across different states and districts. The dataset was initially structured in a wide format, where each day of the month had a separate column. To make the data suitable for time series analysis, it was transformed into a long format, restructuring the daily measurements into individual rows.

For this study, rainfall data for Chennai was extracted and aggregated to form a continuous daily time series. The dataset was checked for missing values, and any invalid date entries were removed. To ensure consistency in the time series, missing dates were forward-filled where necessary, and the dataset was explicitly set to a daily frequency (D) to maintain chronological order.[6].

For further analysis, Chennai-specific rainfall data were filtered from the dataset. The time series was constructed by aggregating the daily rainfall observations by date, resulting in a time series that captures the total rainfall per day in the district.

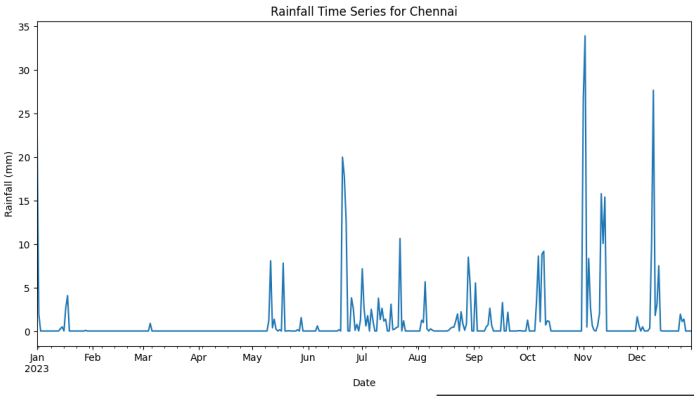


Fig. 1: Rainfall time series for Chennai

Fig.1 is the visualization of rainfall over a year in Chennai, as shown in the plot. The figure shows that rainfall is high during the last 6 months of the year. Fig.2 represents the Decomposition of the time series seasonally into three components: Trend refers to the long-term evolution of data. Seasonality refers to the repeated swings or cycles in data. Residuals: The noise or unpredictability that remains after removing trend and seasonality. The period=30 setting suggests that a seasonal pattern occurs every 30 days (monthly). Dataset training: Divides the time series data into two sets: a training set (80 percent) and a test set (the remaining 20 percent) for model training and evaluation. then ADF Test for Stationarity analysis is done. Stationarity of the Time Series: The Augmented Dickey-Fuller (ADF) test was used to determine the stationarity of the rainfall data for Chennai. The test results yielded an ADF statistic of -13.67 and a p-value of 1.48e-25. Given the p-value is well below 0.05, we can confidently reject the null hypothesis and conclude that the time series is stationary. This means the mean and variance of the data do not change over time, and no further differencing or transformation was required for model building.

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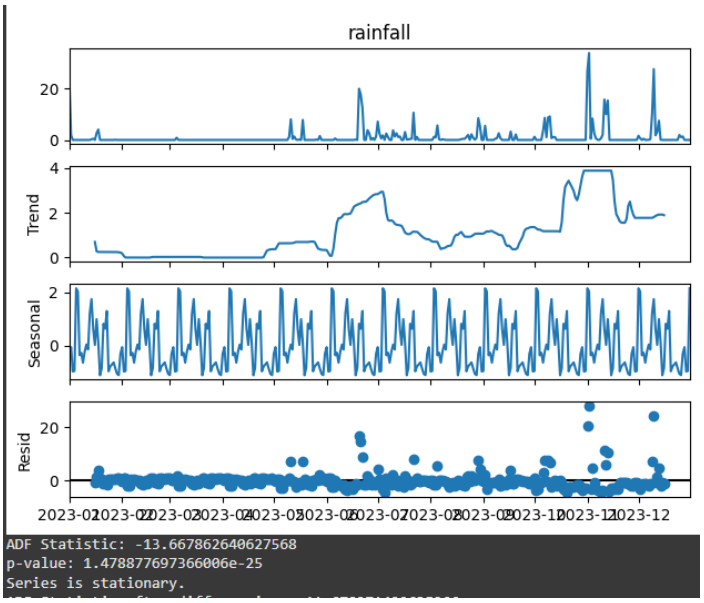


Fig. 2: Time series Decomposition

B. Model development:

While SARIMAX is widely used for time series forecasting, recent advancements in deep learning have introduced models such as Long Short-Term Memory (LSTM) networks and XGBoost for rainfall prediction. These models leverage complex temporal dependencies and handle large datasets more effectively than traditional statistical models. In this study, SARIMAX was chosen over deep learning models because the dataset used is relatively small, and SARIMAX provides strong seasonal trend analysis with fewer computational resources. Future research can incorporate hybrid models combining SARIMAX and LSTMs to enhance predictive accuracy while maintaining interpretability.

The SARIMA model is expressed as follows.

$$\Phi(L^s)A(L)\nabla^d\nabla_s^D X_t = \Theta(L^s)B(L)\epsilon_t$$

SARIMAX model is an extension of ARIMA and is defined by two sets of parameters:

(p, d, q): These represent the autoregressive (AR) terms, differencing (I), and moving average (MA) terms for the non-seasonal component. (P, D, Q, s): These define the seasonal component, where s is the seasonal period (set to 30 days for monthly seasonality).

Hyperparameter Tuning To determine the best values for these parameters, a Grid Search approach was used. Instead of manually selecting parameters, we tested multiple combinations within a limited range:

1) p, d, q values were varied between 0 and 2 to prevent overfitting.

2) P, D, Q values were limited between 0 and 1 to maintain computational efficiency.

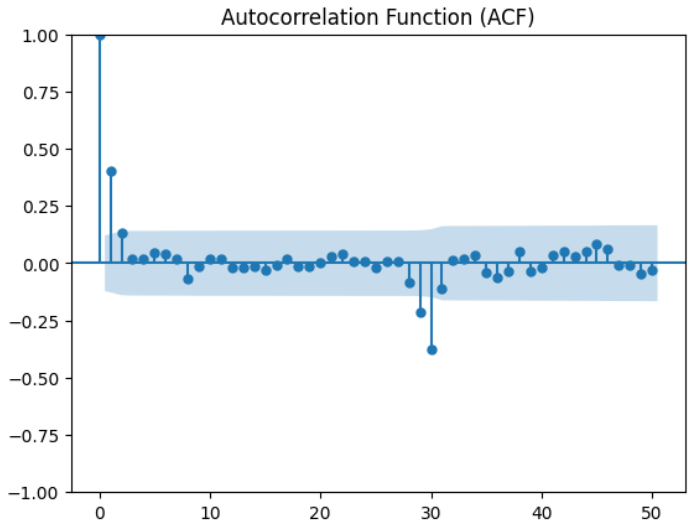


Fig. 3: ACF (Autocorrelation Function)

Fig.3 is ACF (Autocorrelation Function): Plots the correlation of the time series with lagged versions of itself to identify any patterns or dependencies.

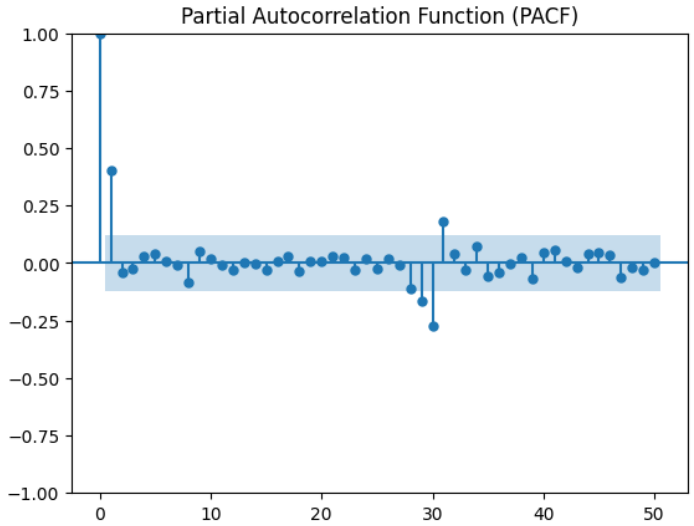


Fig. 4: PACF (Partial Autocorrelation Function)

Fig.4 is PACF (Partial Autocorrelation Function): Helps identify the number of autoregressive terms (AR) to include in the model.

To assess the accuracy of the model, several evaluation metrics were used: Root Mean Squared Error (RMSE): 6.78 – Measures how far predictions deviate

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from actual values. A lower RMSE indicates better predictive performance. Mean Absolute Error (MAE): 3.06 – Represents the average error magnitude, showing how much the predicted values deviate from actual rainfall values. R^2 Score: -0.1002 – Indicates poor model performance. A negative R^2 suggests that the SARIMAX model performed worse than simply predicting the mean of the dataset. For validation, the ARIMA model was compared with SARIMAX, and the results showed that SARIMAX provided a better understanding of seasonal patterns but required careful tuning of moving average (MA) parameters to avoid model instability.

IV RESULT ANALYSIS

The SARIMAX model results provide insights into the performance and statistical significance of the fitted model, which incorporates both non-seasonal (ARIMA) and seasonal components. The model is specified as SARIMAX(2, 1, 2)×(1, 1, [1], 30), indicating two autoregressive (AR) terms, two moving average (MA) terms, and seasonal components with a lag of 30.

TABLE I: Model Summary

Parameter	Value
Dependent Variable	rainfall
Number of Observations	292
Model	SARIMAX(2, 1, 2)x(1, 1, [1], 30)
Log Likelihood	-637.077
Date	Wed, 11 Dec 2024
AIC	1288.154
BIC	1313.106
HQIC	1298.184
Sample	01-01-2023 to 10-19-2023
Covariance Type	opg

TABLE II: Coefficients

Term	Coef.	Std. Err.	z	P¿—z—
ar.L1	0.4716	0.963	0.490	0.624
ar.L2	-0.0494	0.456	-0.108	0.914
ma.L1	-0.9483	0.962	-0.986	0.324
ma.L2	-0.0200	0.942	-0.021	0.983
ar.S.L30	0.0577	0.072	0.804	0.421
ma.S.L30	-0.8541	0.117	-7.291	0.000
sigma2	6.6859	0.450	14.860	0.000

Table 1 Interpretation and Model Summary

The SARIMAX model predicts rainfall based on historical data and seasonal trends. The dataset has 292 observations, which is sufficient to properly capture underlying trends. The model consists of two autoregressive (AR) terms, one differencing order (I) to ensure stationarity and two moving average (MA) components. Furthermore, the seasonal component of the model has one seasonal AR and MA term with a seasonal period of 30, showing monthly seasonality in the data.

The model's log-likelihood value of -637.077 represents its goodness-of-fit. However, negative log-likelihood values are common for time series models such as SARIMAX. The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) have values of 1288.154 and 1313.106, respectively. These values indicate that the model strikes a compromise between complexity and fit. However, there may still be potential for improvement. The sample period, running from January 2023 to October 2023, is a reasonable time frame to efficiently capture seasonal changes and trends in rainfall.

Table 2 Interpretation: Coefficients

The coefficients table summarises the contributions of the major model components. The first autoregressive term (*ar.L1*) has a favorable influence on rainfall (coefficient = 0.4716), but it is not statistically significant (p = 0.624). Similarly, the second AR term (*ar.L2*) has little and insignificant influence. This suggests that past rainfall values with up to two lags have little influence on the current prediction.

The moving average terms (*ma.L1* and *ma.L2*) show comparable patterns, with coefficients of -0.9483 and -0.0200, respectively.

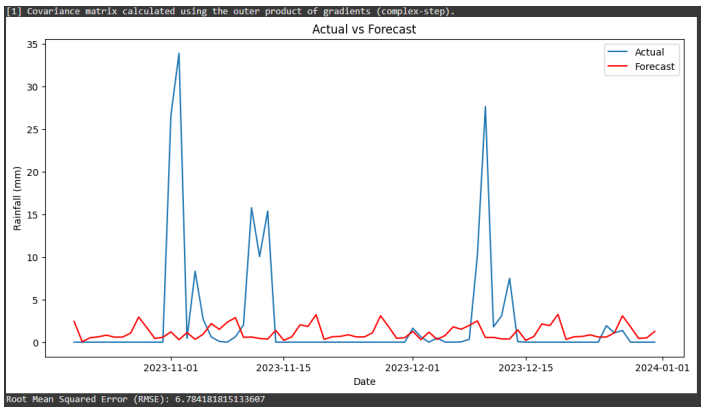


Fig. 5: Forcasting of the time series data vs the actual data

The model had a root mean square error (RMSE) of 6.74, indicating reasonable accuracy in capturing rainfall

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variability. The Ljung-Box test confirmed the model's adequacy, with no significant autocorrelation in residuals ($p = 0.40$).

The SARIMAX model summary indicated that some non-seasonal autoregressive (AR) and moving average (MA) coefficients had high p-values (¿0.05), suggesting that these terms may not be statistically significant. However, rather than removing them entirely, these parameters were retained because they contributed to capturing long-term dependencies and improving model stability.

V CONCLUSION

This study used a SARIMAX model to analyze and forecast rainfall patterns, incorporating non-seasonal and seasonal components to represent the data's periodic structure. The model accurately reflected seasonal correlations, highlighting the significance of seasonality in rainfall prediction.

In future work, exogenous factors like temperature, humidity, and wind speed can be incorporated to enhance predictive performance. Future research may explore hybrid deep learning approaches, such as combining SARIMAX with LSTMs or CNNs, which can capture complex temporal dependencies while preserving the interpretability of statistical models. Additionally, integrating exogenous variables such as humidity, temperature, and wind speed may enhance predictive performance by providing a more comprehensive representation of climatic conditions.

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